

Radxa Rock Pi S0 RK3308B

OS Installation, Serial Setup, and SPI Configuration Documentation

Required Hardware

- Micro SD card ($\geq$ 8GB, Class 10 recommended)
- USB to TTL Serial cable 1.5Mbps (3.3V logic level)
(CP2104 / FT232RL / CH340)
- USB Type-C power cable & adapter (5V/2A recommended)
- Computer with internet access

OS & Downloads Location

Debian Bookworm CLI Image:

[Download Debian Image \(.xz\)](#)

List of Available Images:

<https://dl.radxa.com/rockpis/images/>

Serial Console Setup

Rock Pi S0 GPIO Pinout

Rock Pi S0 GPIO	Signal	Serial Cable Colour
Pin 6	GND	Black wire
Pin 8	TX	White wire (RX)
Pin 10	RX	Green wire (TX)

Warning: Do NOT connect the RED wire.

Serial Configuration Parameters

Parameter	Value
Baud Rate	1500000 (1.5Mbps)
Data Bits	8
Stop Bits	1
Parity	None
Flow Control	None

PuTTY Setup Procedure

1 Identify the COM Port

- Connect the USB to TTL cable to your PC.
- Open *Device Manager* and expand *Ports (COM & LPT)*.
- Note the assigned COM port number (e.g., COM5).

2 Install and Configure PuTTY

- Download PuTTY from the official source: [PuTTY Official Download](#)
- Launch PuTTY and apply the following configuration:
 - **Connection type:** Serial

- **Serial line:** COM5 (or your identified port)
- **Speed:** 1500000
- Click **Open** to establish the terminal connection.

3 Verify Connection

- A blank terminal window will appear. Power on the Rock Pi S0 to observe the boot messages.

Flashing OS Image & Booting

Flashing the OS Image (balenaEtcher)

1. Download & install [balenaEtcher v1.18.11](#).
2. Launch balenaEtcher and click **Flash from file** to select the downloaded image.
3. Click **Select Target** and choose your Micro SD card.
4. Click **Flash!** and wait for the process and verification to complete.

Booting from Micro SD Card

1. Insert the flashed Micro SD card into the Rock Pi S0 card slot.
2. Connect the USB to TTL serial cable to your PC and launch your terminal tool (PuTTY).
3. Connect the USB Type-C power cable (5V/2A).
4. The board will automatically boot from the Micro SD card and display output in the terminal.

Default Credentials & Network Configuration

Default Login Credentials

OS Image	Username	Password
Radxa	radxa	radxa
Radxa (Alt)	rock	Rock
Ubuntu	ubuntu	ubuntu
Debian	debian	debian

WiFi Configuration

Execute the following commands to configure wireless networking:

```
sudo apt update
sudo apt install network-manager # Windows host / Linux
sudo nmcli radio wifi on
sudo nmcli device wifi connect "algofusion" password "OmGanesh"
sudo apt update
sudo apt install build-essential
```

Step-by-Step Procedure to Enable SPI

Step 1: Login to your board via SSH

```
ssh radxa@<board-ip>
# OR
ssh rock@<board-ip>
```

Step 2: Check if SPI device already exists

```
ls /dev/spidev*
```

If no output is returned, proceed with the steps below to enable SPI overlays.

Step 3: Identify boot configuration file

Check the available configuration files under `/boot` :

```
ls /boot
```

Locate either `/boot/config.txt` or `/boot/uEnv.txt` .

Step 4: Enable SPI overlay

If `config.txt` exists, edit the file:

```
sudo nano /boot/config.txt
```

Add the following line:

```
overlays=spi0
```

Note: If `spi0` does not function as expected, update the entry to `overlays=spi1` .

Step 5: Save and Reboot

```
sudo reboot
```

Step 6: Verify SPI device nodes

```
ls /dev/spidev*
```

Expected output:

```
/dev/spidev0.0 /dev/spidev0.1
```

Step 7: Install SPI testing tools

```
sudo apt update  
sudo apt install spi-tools
```

Step 8: Perform SPI Test

```
spidev_test -D /dev/spidev0.0
```

If the test completes without error, SPI has been successfully enabled on your board.